* Digital Communication *

" Computer "

SALEH RAMADAN

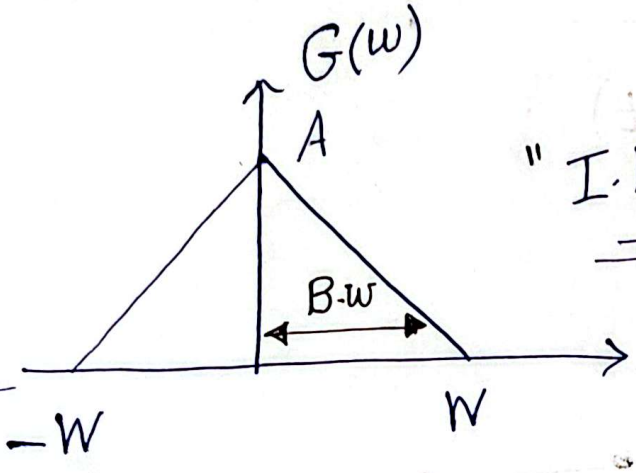
" Sheet 2 - Revision Solution "

*** Problem 1:** $g(t) = \text{SinC}^2(5\pi t)$

i) 5HZ , ii) 10 HZ , iii) 20 HZ

Solution

→ **Remember:**



"I.F.T" $\Longrightarrow g(t) = \underbrace{A * W}_{\text{Area of Traingle}} \text{SinC}^2\left(t * \underbrace{\frac{W}{2}}_{\frac{\text{Total Base}}{4}}\right)$

- In our Problem: We have $g(t) = \text{SinC}^2(5\pi t)$

Then: $\frac{W}{2} = 5\pi \Longrightarrow W = 10\pi \Longrightarrow w_m = 10\pi$

$f_m = 5$ HZ $\xrightarrow{\text{Nyquist}}$ $f_s \geq 2f_m$

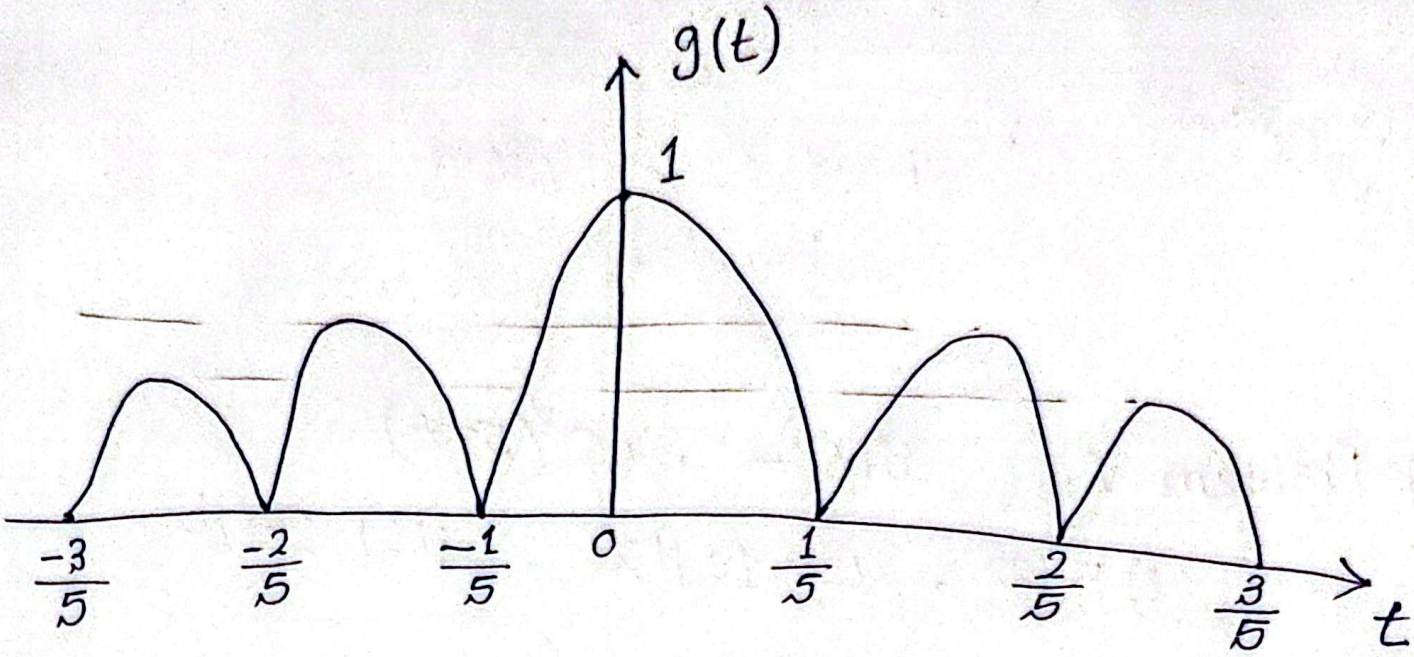
$\Downarrow$

To Avoid Aliasing ← $f_s \geq 10$

CamScanner

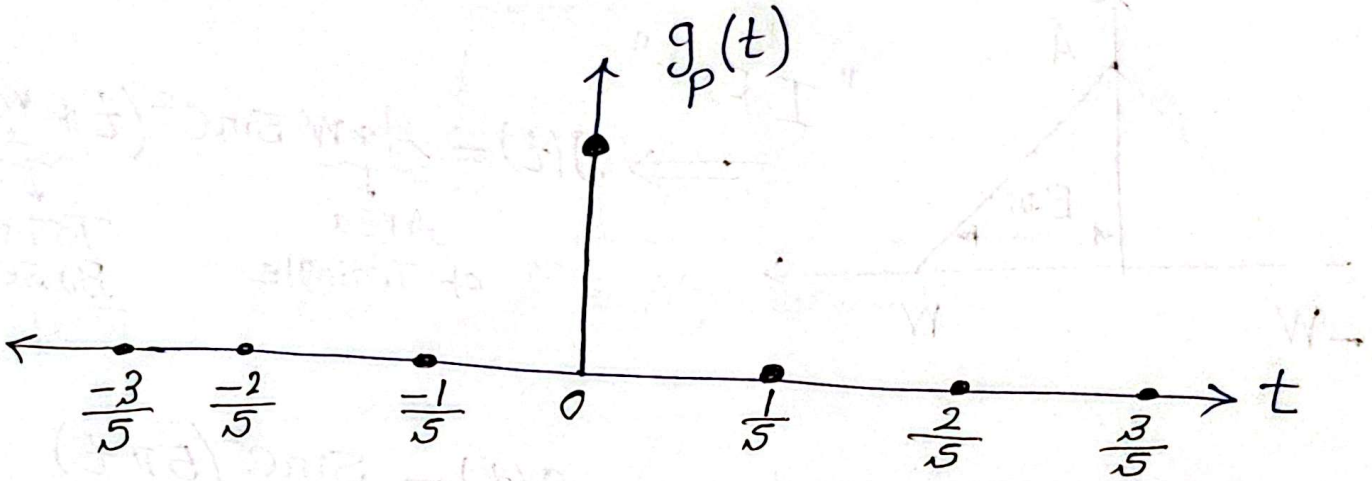
②

• Zeros of $Sinc^2$: $5\pi t = n\pi \Longrightarrow \boxed{t = \frac{n}{5}}$



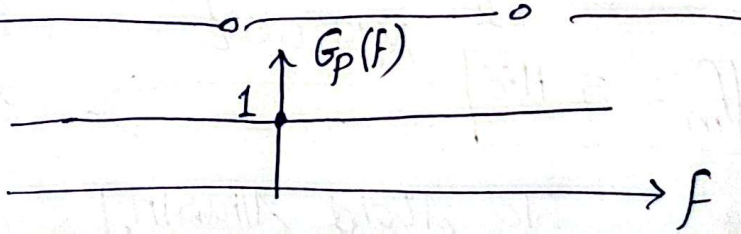
i $f_s = 5Hz \Longrightarrow f_s < 10\ Hz \Longrightarrow$ Aliasing

a) $T_s = \frac{1}{f_s} \Longrightarrow \boxed{T_s = \frac{1}{5}}$



Then: $\boxed{g_p(t) = \delta(t)}$

b) $G_p(f) = 1 \Longrightarrow$



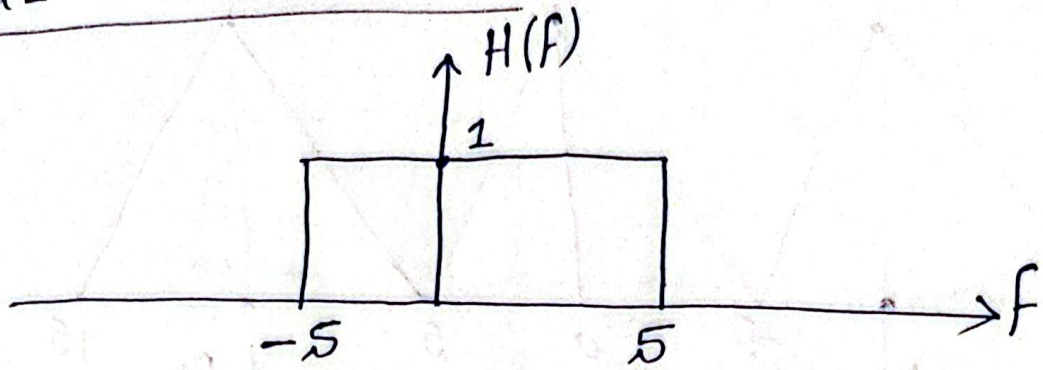
CamScanner

c) We can not recover signal, as

$f_s <$ Nyquist Rate.

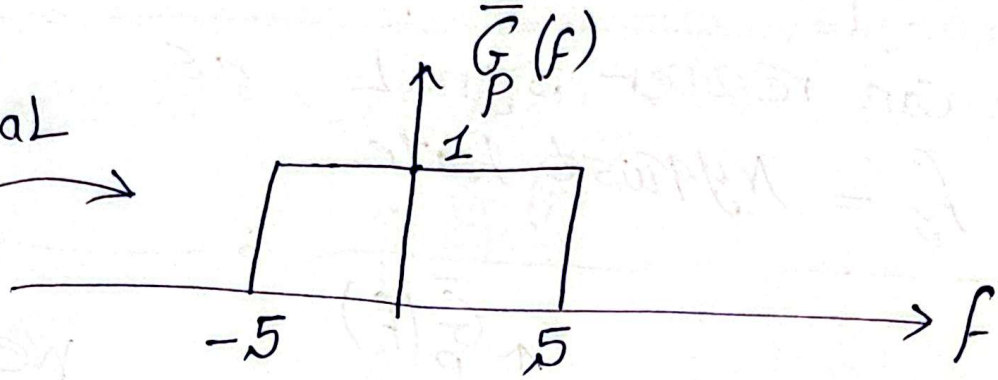
d)

. Ideal Low-Pass Filter:



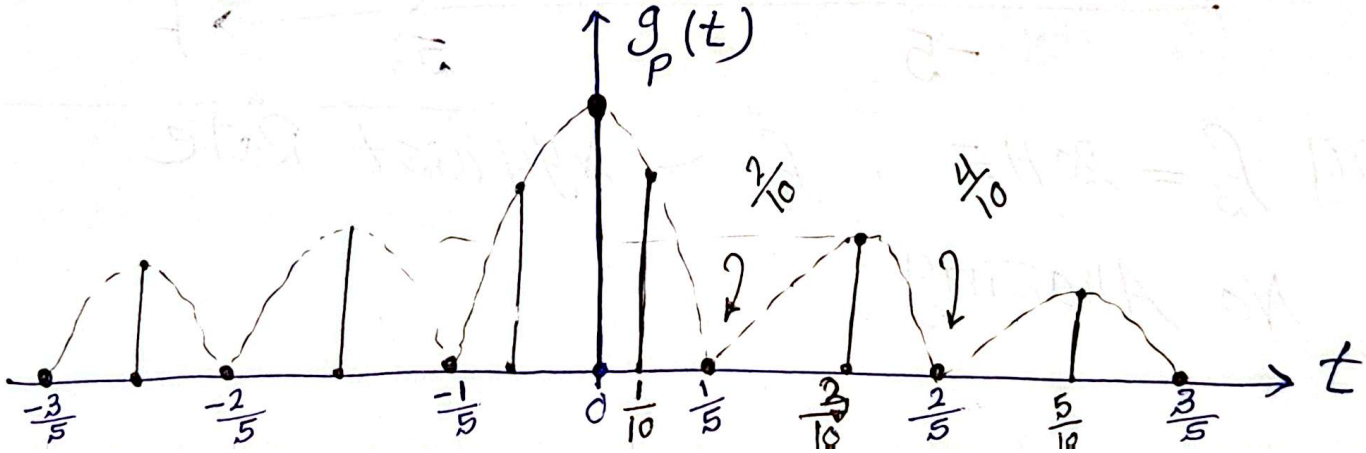
$\bar{G}_p(f)$: $G_p(f)$ after Passing LPF

It's not The signal



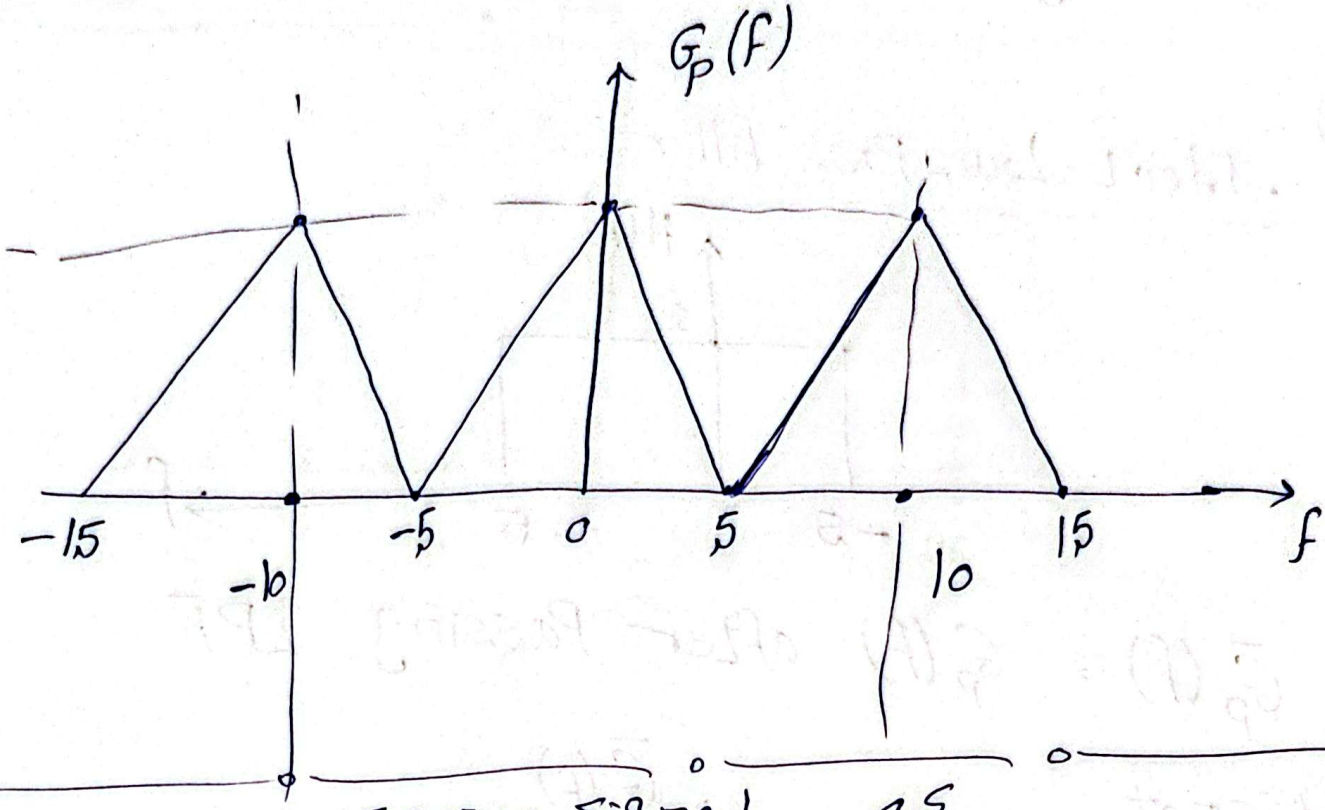
[ii] $f_s = 10\,Hz$, $f_s =$ Nyquist Rate $\Rightarrow$ No-Aliasing

a) $T_s = \frac{1}{f_s} \Longrightarrow T_s = \frac{1}{10} \Longrightarrow T_s = \frac{1}{5} * \frac{1}{2}$



④

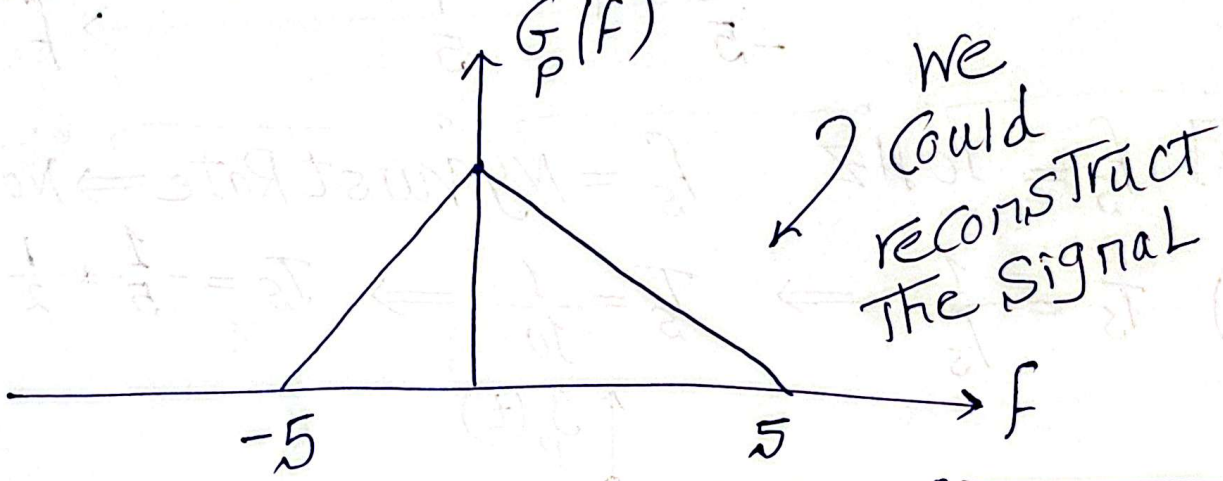
b) Replicate shape of the spectrum of the signal each f_s



c) we can recover signal, as

f_s = Nyquist Rate

d)

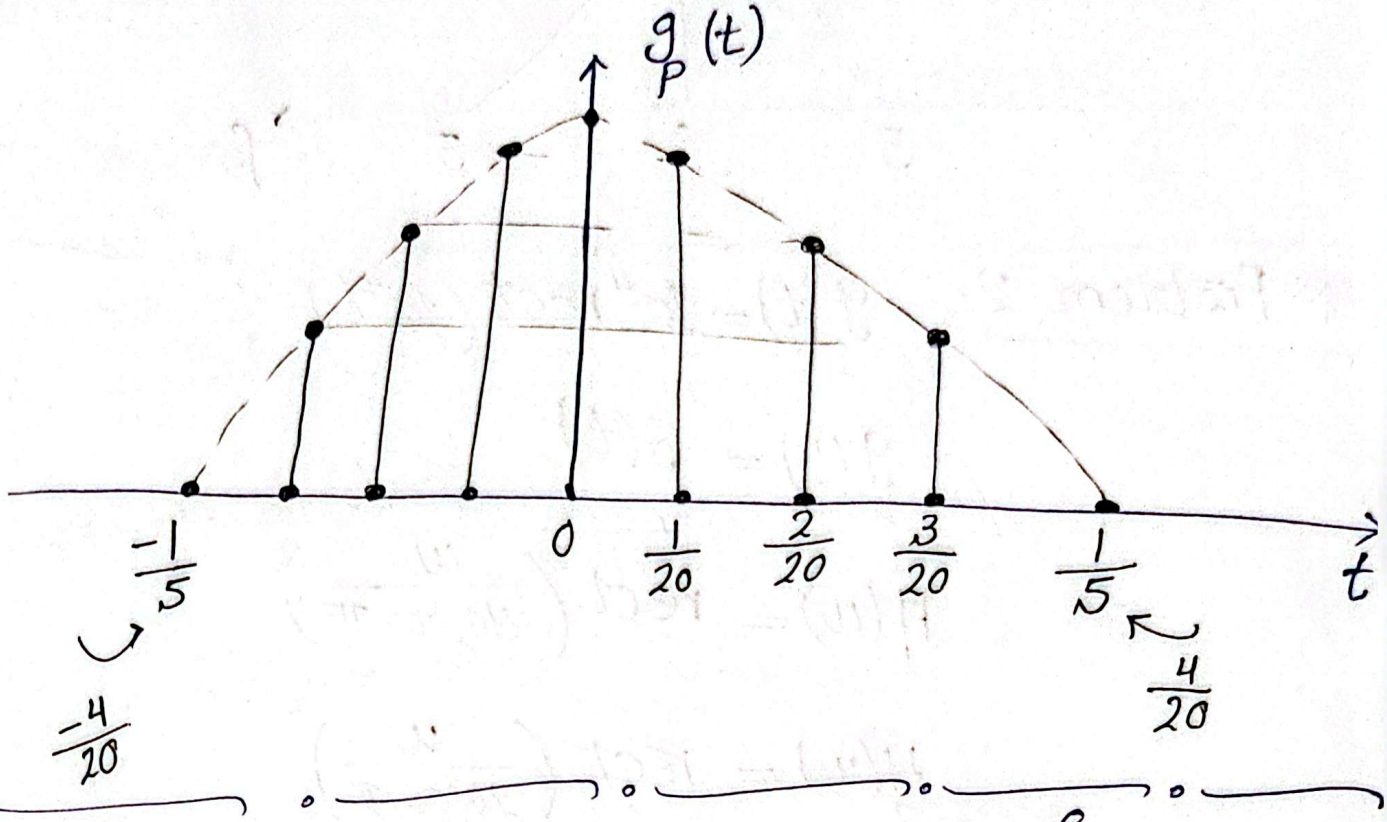


We could reconstruct the signal

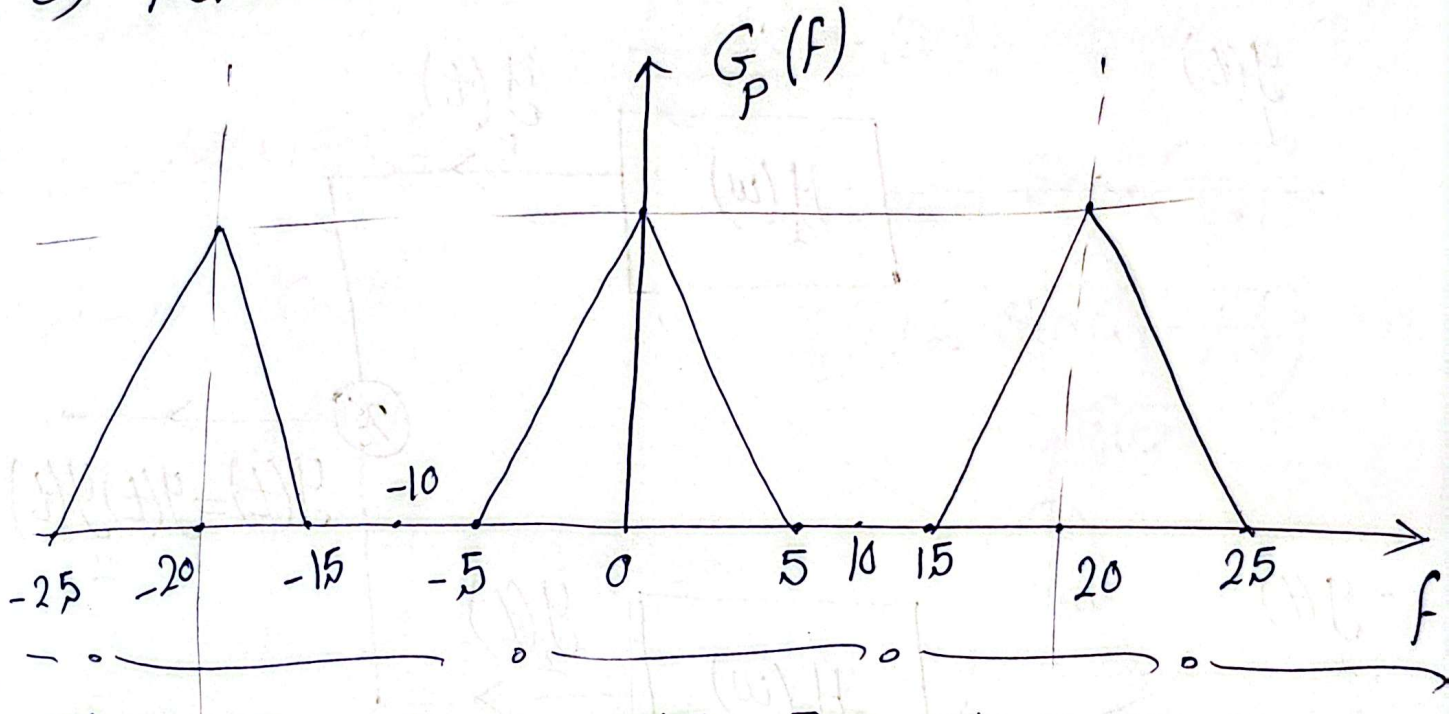
iii) $f_s = 20\,Hz$, $f_s >$ Nyquist Rate

No Aliasing.

a) $T_s = \frac{1}{f_s} \Longrightarrow T_s = \frac{1}{20}$, $T_s = \frac{1}{5} * \frac{1}{4}$



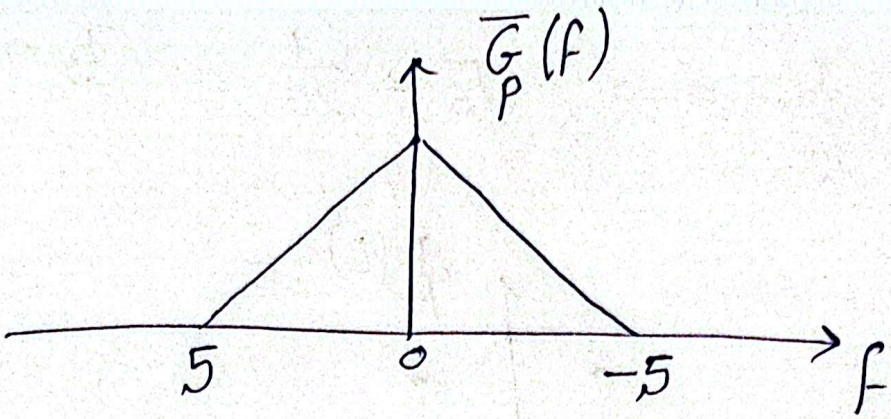
b) Replicate The spectrum each f_s



c) we can recover the signal, as

$f_s >$ Nyquist Rate.

d)

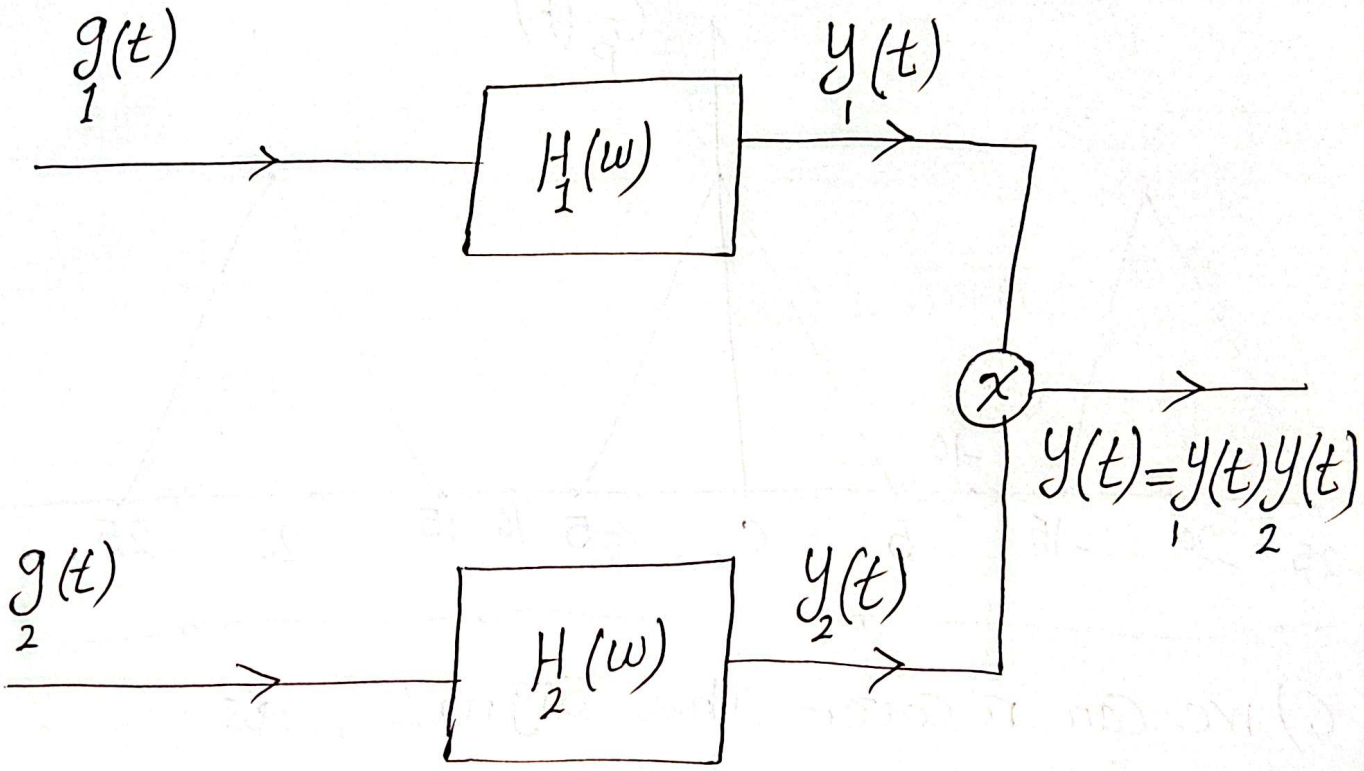


* Problem 2: $g_1(t) = 10^4 \, \text{rect}(10^4 t)$

$g_2(t) = \delta(t)$

$H_1(w) = \text{rect}\left(\frac{w}{40,000\pi}\right)$

$H_2(w) = \text{rect}\left(\frac{w}{20,000\pi}\right)$



Solution

To get Nyquist Rate, we need to get B.W of Messages $y_1(t)$, $y_2(t)$, $y(t)$

So we need their spectrums

→ For $y_1(t)$:

$$y_1(t) = g_1(t) \circledast h_1(t)$$

$$Y_1(f) = G_1(f) * H_1(f)$$

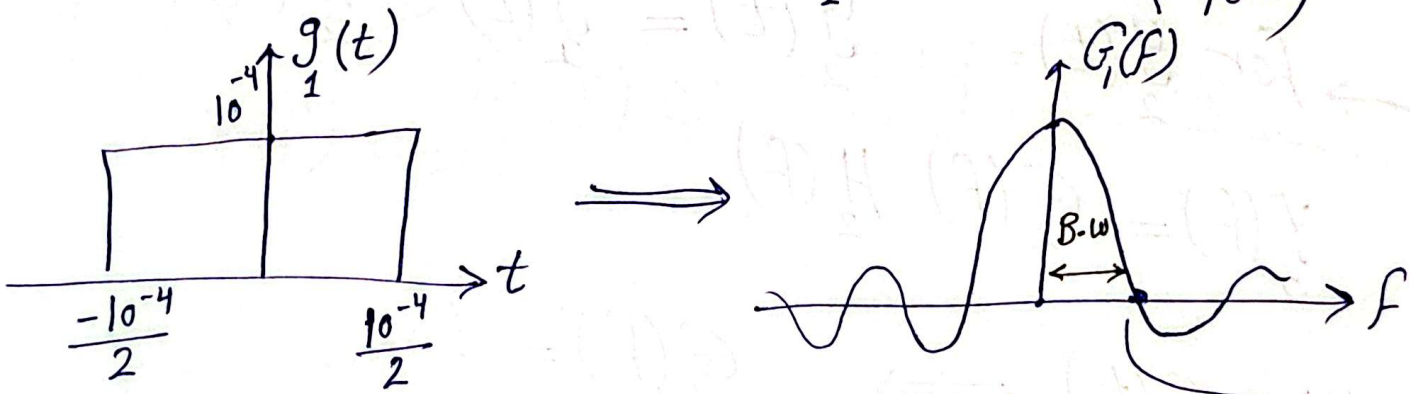
$$g_1(t) = 10^4 \, rect(10^4 t) \xrightarrow{F.T} G_1(w) = A*\tau \, Sinc\left(\frac{w*\tau}{2}\right)$$

$$g_1(t) = A * rect\left(\frac{t}{\tau}\right)$$

$$G_1(w) = Sinc\left(\frac{w}{2*10^4}\right)$$

$A = 10^4$, $\tau = 10^{-4}$

$$G_1(f) = Sinc\left(\frac{\pi f}{10^4}\right)$$

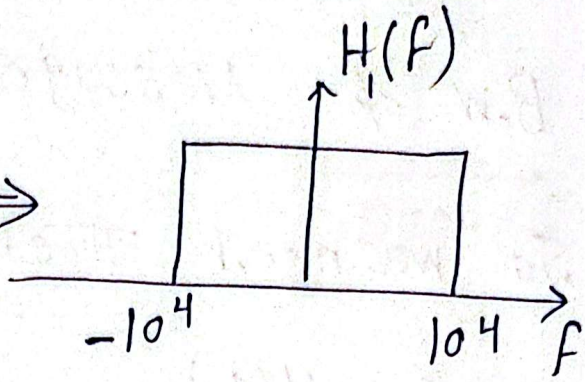


Zeros of Sinc: $\frac{\pi f}{10^4} = n\pi$, @ $n = 1$

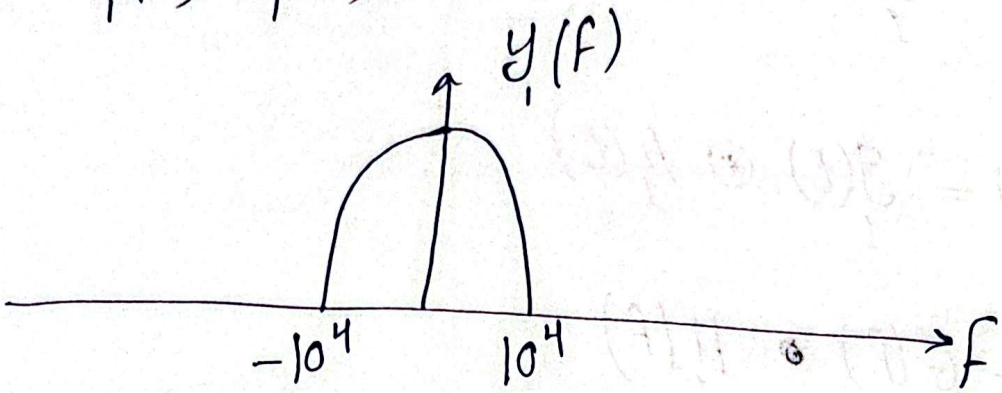
$f = 10^4$

$$H_1(\omega) = \text{rect}\left(\frac{\omega}{40{,}000\pi}\right)$$

$$H_1(f) = \text{rect}\left(\frac{f}{20{,}000}\right) \Longrightarrow$$



$$Y_1(f) = G_1(f)\, H_1(f)$$



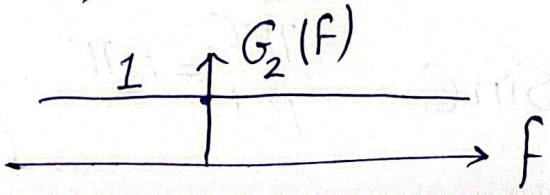
$$f_{y_1} = B_{y_1}\text{-}w = 10^4 \quad , \quad f_{S_{y_1}} = 2\, B_{y_1}\text{-}w$$

$$\boxed{f_{S_{y_1}} = 2 * 10^4 \text{ HZ}} \Longrightarrow$$

→ for $y_2(t)$: $\quad y_2(t) = g_2(t) \circledast h_2(t)$

$$Y_2(f) = G_2(f)\, H_2(f)$$

$$g_2(t) = \delta(t) \Longrightarrow \quad G_2(f) = 1$$

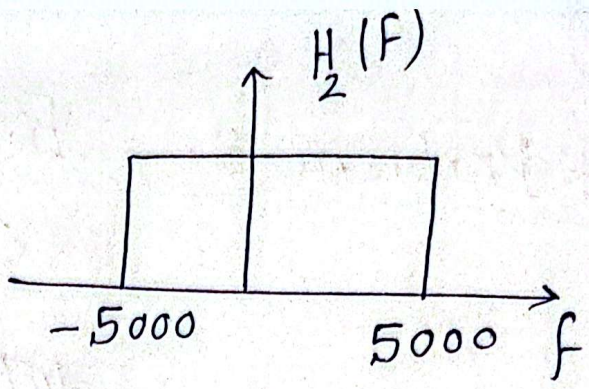


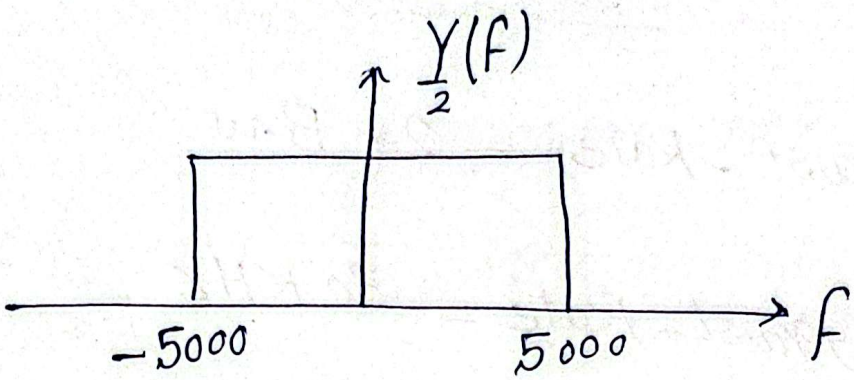
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(9)

$$H_2(\omega) = \text{rect}\left(\frac{\omega}{20{,}000\pi}\right)$$

$$H_2(f) = \text{rect}\left(\frac{f}{10{,}000}\right)$$





$$B.w\Big|_{y_2} = 5000 \Longrightarrow \boxed{f_s\Big|_{y_2} = 10{,}000\ Hz}$$ #

→ For $y(t)$: $\quad y(t) = y_1(t)\ y_2(t)$

Multiplication in Time $\Longrightarrow$ Convolution in Frequency

$$Y(f) = \frac{1}{2\pi}\left[Y_1(f) \circledast Y_2(f)\right]$$

$$B.w\Big|_{y} = B.w\Big)_{y_1} + B.w\Big)_{y_2} \Longrightarrow B.w\Big)_{y} = 15{,}000$$

$$\boxed{f_s|_y = 30{,}000\ Hz}$$ #

* Problem 3:

B.w) Audio = 15 KHZ

Solution

a) Nyqwist - Rate = 2 * B.w

Nyqwist_Rate = 30 KHZ ✻

b) $L = 65,536$, $n = \lceil \log_2(L) \rceil$

n = 16 bits/sample ✻

c) $R_b = n * f_s$, f_s = 30 K sample/sec

R_b ← bits/sample ← n ; f_s ← sample/sec

bits/sec.

R_b = 480 KbPS ✻

d) $f_s|_{New}$ = 44,100 sample/sec. , $L = 65,536$

$R_b|_{New} = n * f_s|_{new} \Longrightarrow$ **$R_b|_{New}$ = 705.6 KbPS.** ✻

$B.W)_{min} = 0.5 R_b \Longrightarrow \boxed{B.W)_{min} = 3528 \text{ KHz}}$ ✻

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*Problem 4: $B.W_{ECG} = 100 \text{ Hz}$

$$\text{Max-error} = \frac{0.25}{100} A$$

$$f_s = 2 * \text{Nyquist_Rate}$$

$B.W)_{min} = 0.5 R_b$

$$R_b = \frac{\text{bits}}{\text{Time}} = \frac{n * N}{T_s} = N * n * f_s$$

Solution

$$R_b = \#\text{Signals} * n * f_s = a f_x$$

R_b: Bits/sec For TDM Frame

#Signals: 10

$f_x = \#\text{Signals in TDM Frame} * f_s$

Get f_s: $\text{Nyquist_Rate} = 2 * B.W_{ECG}$

$\text{Nyquist_Rate} = 200 \text{ Hz}$

$\boxed{f_s = 400 \text{ Hz}}$

Get n: $\text{Max-Error} = \frac{\Delta}{2} = \frac{0.25}{100} A$

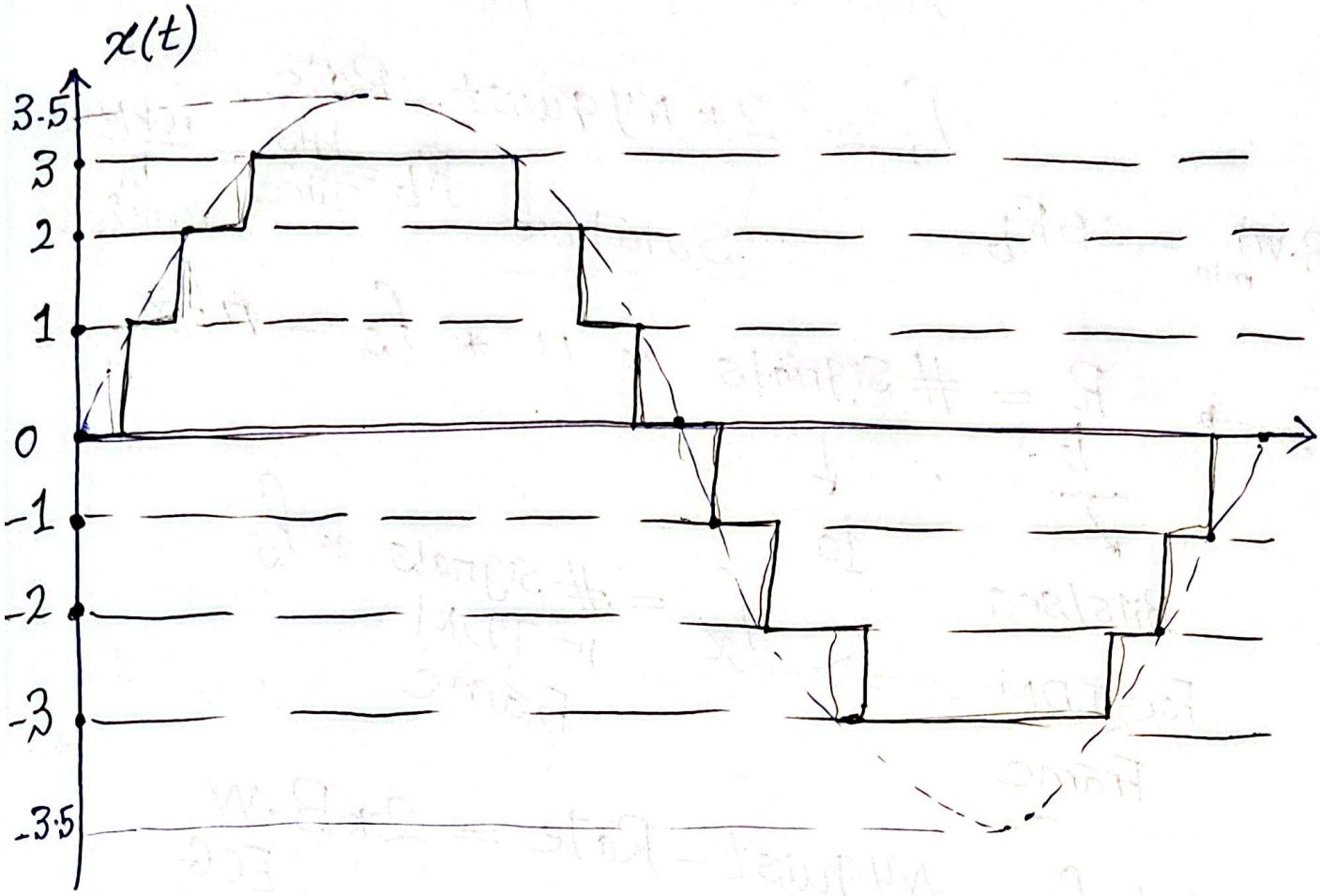
$\Delta = \frac{2A}{L} \Longrightarrow L = 400 \Longrightarrow n = \lceil \log_2(L) \rceil$

$\boxed{n = 9 \text{ bits/sample}} \Longrightarrow R_b = 36 \text{ KbPS} \Longrightarrow \boxed{B.W_{min} = 18 \text{ KHz}}$ ✻

* Problem 5: $x(t) = 3.25 \sin(t)$

*** Case 1:** Mid-Thread : Zero is a Level

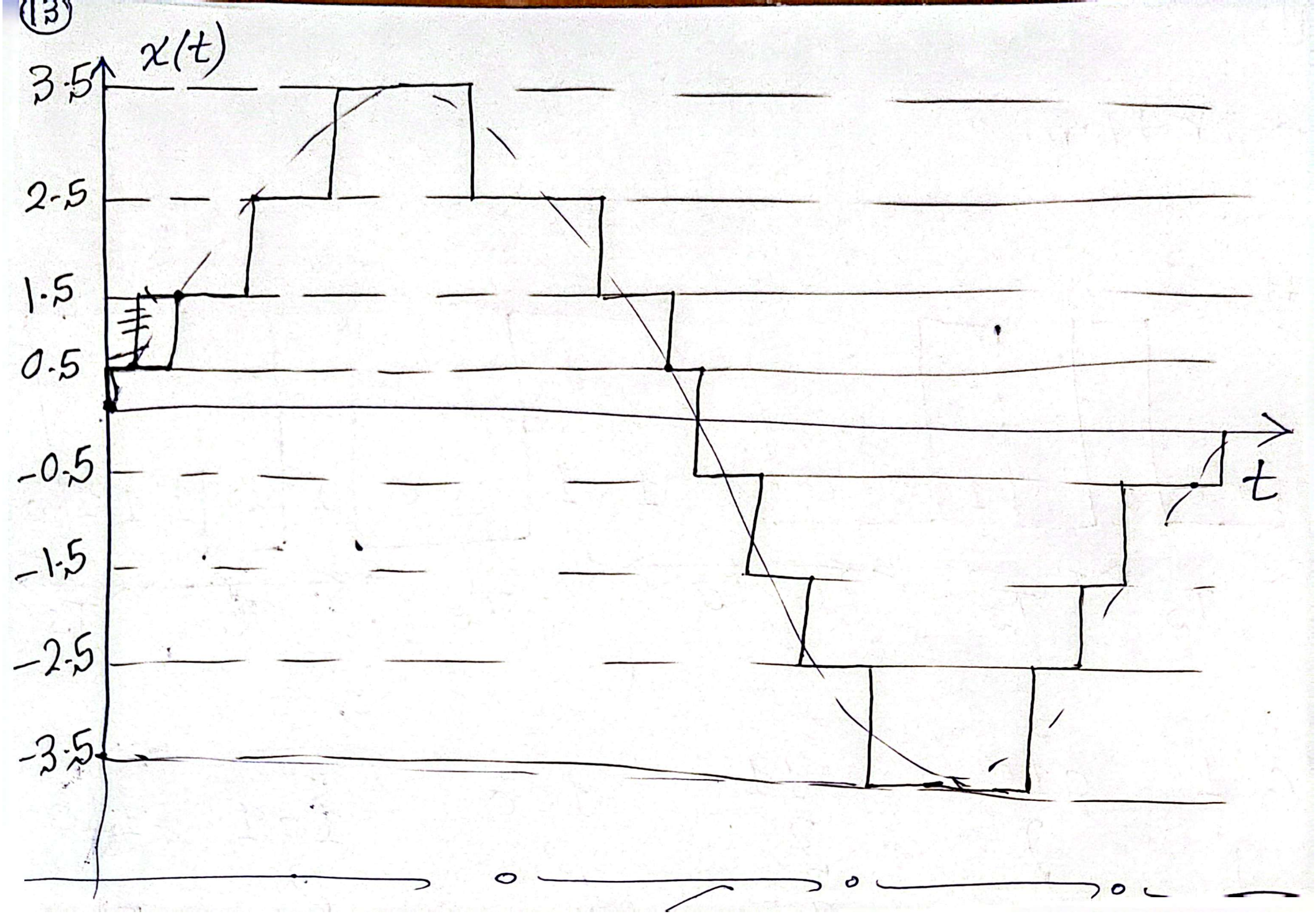
Levels: 0, ±1, ±2, ±3 (7-Levels)



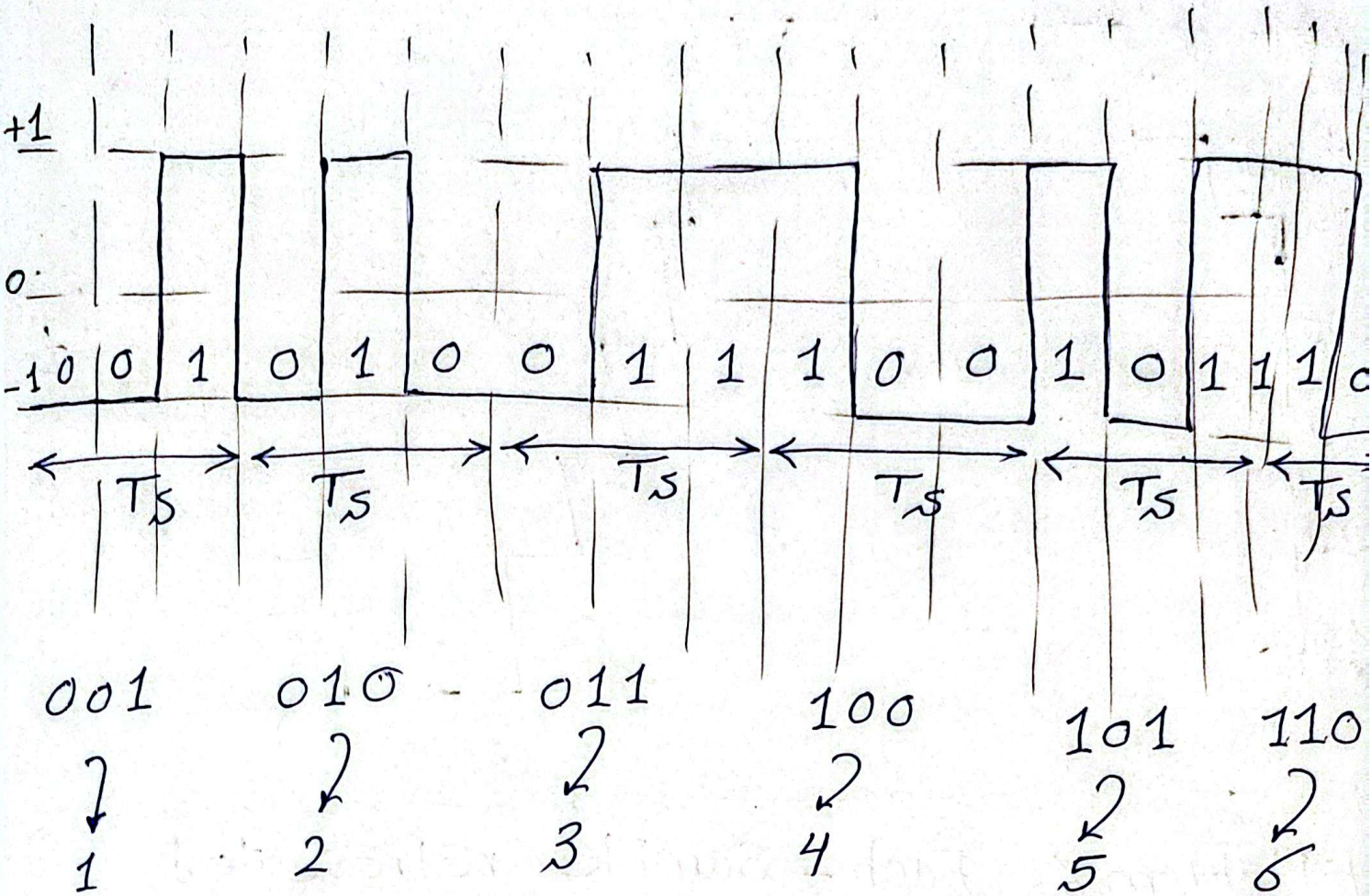
*** Case 2:** Mid-Rise: Zero is not a Level

Levels: ±0.5, ±1.5, ±2.5, ±3.5 (8-Levels)

(13)



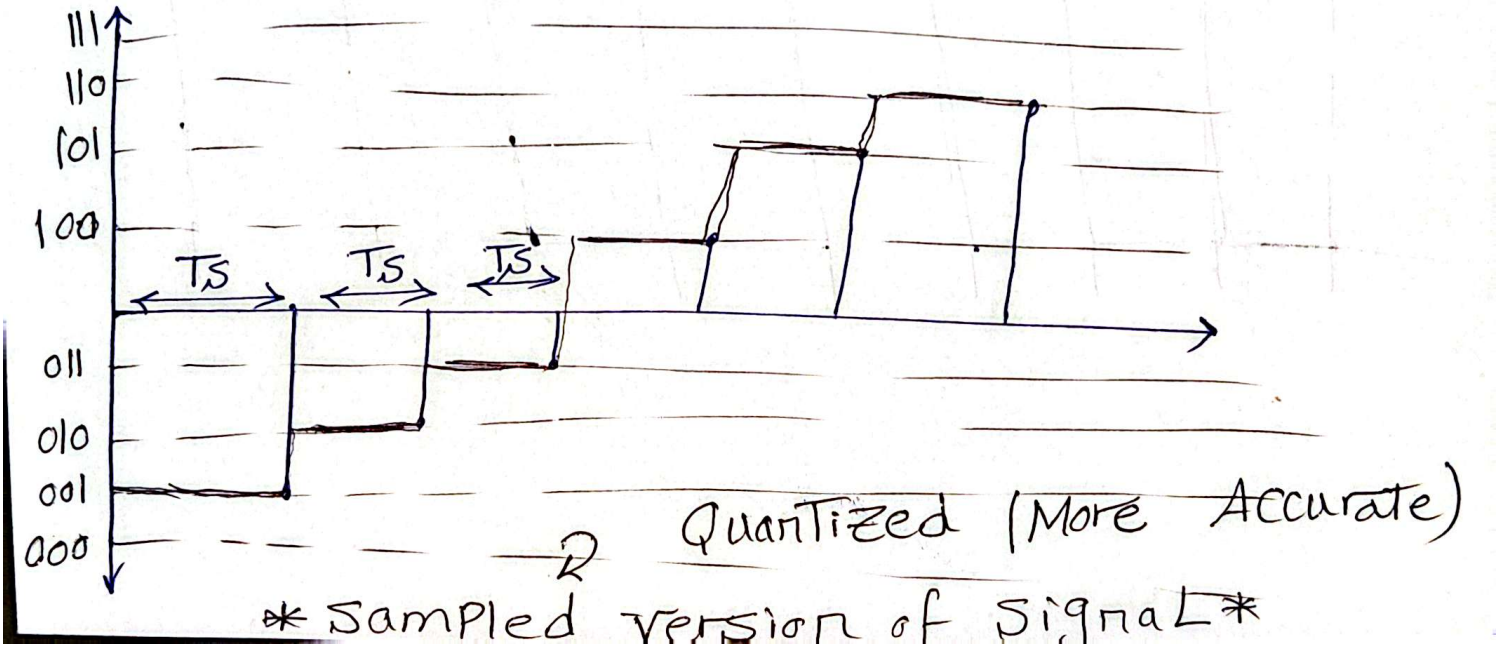
*Problem 6:



001 010 011 100 101 110

1 2 3 4 5 6

$n=3 \Longrightarrow L=8 \quad \therefore T_s = 3T_b$



Quantized (More Accurate)

2

Sampled version of signal